Predict-Then-Commit Speaker Notes

Slide-by-slide talking points for the cleaned end-term deck

How to use these notes

- Treat each slide as a 30-60 second narration block unless it is marked as an appendix.
- Do not read the slide verbatim. Use the slide as visual evidence and these notes as your spoken story.
- Main talk can stop at Slide 37. Slides 38-41 are backup for technical questions.
- If any projected result is not actually run yet, present it as rehearsal/future-state only; for formal academic claims, use verified experimental numbers.

Slide 1: Title

What is on the slide: Title slide with project name, subtitle, author, university, and end-term framing.

What to explain:

- Open with one clean sentence: 'This project is about making streaming translation not only accurate and fast, but stable enough for a human to read in real time.'
- Mention that standard MT assumes full context, while streaming MT must speak before the sentence is complete.
- Frame the contribution around stable prefix commitment: the system learns when a partial translation is safe to show permanently.
- Transition: 'To see why this matters, let me start with the core claim of the project.'

***Speaker cue:** Calm, confident, research-focused opening. Do not apologize for complexity.*

Slide 2: Executive Claim

What is on the slide: Main thesis box, three mini-claims: problem, missing metric, solution, and target of preserving BLEU while driving visible retraction to zero.

What to explain:

- State the main thesis directly: existing streaming MT evaluation usually tracks quality and latency, but users also experience visible instability.
- Explain visible text instability in simple terms: text appears on screen, then changes or disappears as more source context arrives.
- Clarify the gap: token-level lag or BLEU alone cannot measure the annoyance of text being visually rewritten.
- Say the solution: instead of decoding as early as possible, we learn when to commit a stable prefix.
- End with the target: keep translation quality high while eliminating visible retraction.

***Speaker cue:** This is your elevator pitch. Make it sound like the whole project can be summarized by one missing objective.*

Slide 3: Roadmap

What is on the slide: Seven-part outline: problem, formalization, metrics, method, evaluation, results, ablations/failures/takeaways.

What to explain:

- Walk through the structure quickly. Do not read every line slowly like a hostage note.
- Say the talk first motivates the failure mode, then introduces the formal objects, then shows metrics and method.
- Emphasize that the second half is about corrected evaluation: proper BLEU/chrF/COMET, true autoregressive baselines, and logged prefix-level evidence.
- Transition: 'Let's begin with the failure that creates the whole problem.'

***Speaker cue:** Keep this under 25 seconds.*

Slide 4: The Core Failure: Streaming Decoding Under Uncertainty

What is on the slide: Side-by-side diagram comparing offline NMT with full context versus streaming NMT with only source prefix and possible visible revision.

What to explain:

- Explain offline NMT first: the model receives the complete source sentence before generating, so ambiguity can often be resolved before output.
- Then explain streaming NMT: the model sees only a prefix, so it may generate a plausible but wrong translation.
- The important sentence: 'A wrong early prediction is not just a model error; if shown to the user, it becomes a visible retraction event.'
- This sets up why stability is a separate user-facing objective.

Speaker cue: *Point to both columns and contrast 'full context' vs 'context arrives after speaking.'*

Slide 5: Running Example: The bank will raise rates

What is on the slide: Timeline showing source stream, retranslation behavior, and commit-policy behavior for the ambiguous word 'bank.'

What to explain:

- Use this as the intuitive example of the entire project.
- Say that 'bank' is ambiguous early: it can mean riverbank or financial bank.
- Retranslation may first show the wrong sense, then revise when 'raise rates' appears.
- Our commit policy is allowed to internally change its mind, but it delays what the user sees until the phrase is stable.
- Core line: 'Internal uncertainty is acceptable; visible instability is the problem.'

Speaker cue: *This slide should feel practical, not mathematical.*

Slide 6: The Three-Way Tradeoff

What is on the slide: Triangle plot showing quality, low latency, and stability; Immediate, Wait-k, Local agreement, Predict-Then-Commit, and offline upper bound.

What to explain:

- Explain that no streaming system gets all three for free.
- Immediate output is low latency, but it flickers.
- Offline full-sentence translation has high quality and stability but is not truly streaming.
- Wait-k and local agreement sit in the middle with fixed or heuristic behavior.
- Predict-Then-Commit tries to move toward the Pareto frontier by learning the quality-latency-stability tradeoff explicitly.

Speaker cue: *This is the conceptual map for the whole deck.*

Slide 7: Three Objects Reviewers Need to See

What is on the slide: Three boxes: tentative hypothesis h(t), committed prefix c(t), visible output z(t), plus core correction from midterm.

What to explain:

- Say this is the key formal correction from the midterm version.
- Define h(t): the internal decoder guess. It is allowed to change.
- Define c(t): the stable prefix that has been locked and should only grow.
- Define z(t): the exact string shown to the user; flicker exists only here.

- Emphasize that metrics must be computed on z(t), not merely internal hypotheses.

Speaker cue: *This is one of the most important slides. Slow down slightly.*

Slide 8: Streaming Setup

What is on the slide: Mathematical setup: source sentence x, observed prefix x_{<=g_t}, monotone read schedule, and final reference y*.

What to explain:

- Introduce notation only as much as needed.
- x is the source token sequence; at time t, the system has read only up to g_t.
- The read schedule is monotone: we can read more source tokens, but we cannot unread them.
- y* is the final reference used for evaluation.
- Transition: 'Now that the streaming setup is fixed, we can define what the model thinks internally.'

Speaker cue: *Do not over-explain the symbols; the audience only needs the roles.*

Slide 9: Tentative Hypothesis h(t)

What is on the slide: Equation defining h(t) as argmax translation under current source prefix and previous committed prefix; interpretation and non-monotonicity warning.

What to explain:

- Say h(t) is the model's best current guess under partial context.
- It is conditioned on the source prefix and the already committed prefix.
- Important: h(t) is internal. It is not necessarily shown to the user.
- The sequence of hypotheses is not monotone; h(t) may not be a prefix of h(t+1).
- This non-monotonicity is where uncertainty lives.

Speaker cue: *Phrase to repeat: 'h(t) can change; z(t) should not flicker.'*

Slide 10: Committed Prefix c(t)

What is on the slide: Definition of committed prefix, append-only monotonicity, concatenation with delta c(t), and compatibility with current hypothesis.

What to explain:

- Define c(t) as the prefix we have decided to expose as stable.
- The key constraint is append-only monotonicity: once committed, text cannot be deleted.
- The delta term is simply the newly added committed part.
- The compatibility constraint says the committed prefix should be consistent with the current tentative hypothesis.
- This is the formal version of 'only show what we are willing to stand behind.'

Speaker cue: *This slide gives your method its backbone.*

Slide 11: Visible Output Stream z(t)

What is on the slide: Detokenization D(.), committed-only display z(t)=D(c(t)), tentative-preview display z(t)=D(c(t)⊕u(t)), and evaluation principle.

What to explain:

- Define z(t) as the surface string the user actually sees.
- In committed-only mode, z(t) is just the detokenized committed prefix.
- If a system displays tentative preview text, z(t) includes an uncommitted suffix that may change later.
- The evaluation principle is the key: all stability metrics are computed over visible output states, not hidden model states.
- This avoids claiming stability just because internal chaos is hidden incorrectly.

Speaker cue: *Connect this back to slide 7.*

Slide 12: Metric 1: Detokenized Erasure

What is on the slide: Definition of LCP, step-wise visible erasure, total DE, normalized DE, and reason for character-level evaluation.

What to explain:

- Explain LCP as the longest common prefix between two consecutive visible strings.
- If the previous string had characters beyond the new LCP, those characters had to be retracted.
- Summing this over time gives Detokenized Erasure.
- Normalized DE makes scores comparable across sentence lengths.
- Why character-level? Because users see characters, not BPE tokens. Subword token edits can exaggerate or hide real visual change.

Speaker cue: *This is a metric contribution. Present it as one of your original pieces.*

Slide 13: Metric 2: Character-Normalized Average Lagging

What is on the slide: Definition of token character mass, source mass, visible target mass, gamma ratio, instantaneous lag, and corpus CNAL.

What to explain:

- Start with the problem: token-level latency can be misleading when source and target languages segment information differently.
- $\chi(v)$ gives the visible character mass of a token after stripping BPE markers.
- Csrc is how much source character mass has been read; Cvis is how much target character mass has been shown.
- Gamma corrects average source-target length ratio.
- CNAL penalizes the system when it has read much more information than it has visibly emitted.
- Use English-to-German compounding as the motivating example.

Speaker cue: *Say CNAL is complementary to AL/DAL, not a replacement for all latency metrics.*

Slide 14: Metric 3: Commit Delay

What is on the slide: Anti-silence metric: token-level commit delay and character-weighted commit delay.

What to explain:

- Open with the danger: a model can get zero erasure by saying nothing until the end.
- Commit delay prevents that cheat.
- r(i) is the source read position when target token i first appears in the committed prefix.
- Dcommit averages this over tokens; Dchar weights by character mass.
- This forces the system to be stable without becoming silent.

Speaker cue: *This slide answers the obvious reviewer attack: 'Is zero erasure trivial?'*

Slide 15: Metric Suite Used in Final Evaluation

What is on the slide: Table of quality, latency, stability, commit behavior, and runtime metrics; bottom note that all baselines use true autoregressive decoding.

What to explain:

- Tell the audience the final evaluation is deliberately multi-objective.
- Quality is measured by SacreBLEU, chrF++, and COMET.
- Latency uses standard AL/DAL and the new CNAL.
- Stability uses DE/NDE, while commit delay prevents silence.
- Runtime measures deployment feasibility.
- Emphasize the correction: all baselines now use true autoregressive decoding, not shortcuts.

Speaker cue: *This slide shows the evaluation is reviewer-aware.*

Slide 16: System Overview

What is on the slide: Pipeline diagram: source stream, BPE prefix, masked encoder memory, autoregressive decoder, commit policy, committed output, and reward.

What to explain:

- Walk through the diagram left to right.
- The source arrives as a stream and is converted into BPE prefixes.
- The encoder memory is masked so the decoder only uses available source context.
- The autoregressive decoder produces candidate output, while the commit policy decides READ or COMMIT.
- The reward combines quality, erasure, CNAL, and commit delay.
- Important: the policy controls exposure of output, not just translation generation.

Speaker cue: *Use this as the bridge from metrics to method.*

Slide 17: Policy State and Action Space

What is on the slide: State vector with decoder state, entropy/confidence, read position, committed length, source mass, visible mass; actions READ/COMMIT; softmax MLP policy.

What to explain:

- Define the policy state in plain English: it knows what the decoder believes, how confident it is, how much source has been read, and how much output has been shown.
- The action space is intentionally simple: READ more source or COMMIT stable output.
- READ advances the source prefix; COMMIT appends a stable target token or span.
- The policy is a lightweight MLP, not a heavy second translation model.
- This modularity matters because it makes the method easier to train and deploy.

Speaker cue: *Do not get trapped explaining every dimension of the vector.*

Slide 18: Training: No More Actor-Critic Claim

What is on the slide: Terminology correction, Stage 1 MLE warm-start equation, Stage 2 REINFORCE policy-gradient update with moving-average baseline.

What to explain:

- Say this slide is an intentional correction and strengthening from the earlier version.
- The implemented method is policy-gradient fine-tuning with a moving-average baseline, not actor-critic unless we add a learned value function.
- Stage 1 trains the translation backbone with MLE under a wait-k streaming mask.
- Stage 2 freezes the translator and trains only the commit policy using REINFORCE.
- This prevents catastrophic forgetting and makes the claim technically accurate.

Speaker cue: *This honesty makes the work stronger. Reviewers like precision, because apparently that is their entire personality.*

Slide 19: Algorithm: Stable Prefix Commitment

What is on the slide: Eight-step algorithm from reading source prefix to optimizing reward at sentence end.

What to explain:

- Present this as the operational version of the method.
- First, read a source prefix and generate a tentative hypothesis autoregressively.
- Then compute the policy state and choose READ or COMMIT.
- READ updates source mass; COMMIT updates committed output and visible string.
- At sentence end, the reward evaluates quality, erasure, latency, and delay.
- This algorithm turns visible stability into an optimization problem.

Speaker cue: *Good place to pause before experiments.*

Slide 20: Experimental Design

What is on the slide: Datasets, models, baselines, and fairness rule.

What to explain:

- Explain that the design evaluates both news-style and TED-style English-German streaming.
- Mention the stress subset: compounds, verb-final syntax, and long clauses, because these are hard for streaming German translation.
- List baselines: offline upper bound, immediate retranslation, wait-k variants, local agreement, confidence-threshold commit.
- The fairness rule is the most important line: same tokenizer, same test set, same decoding budget, true autoregressive decoding.
- This avoids weak-baseline criticism.

Speaker cue: *This is the defense against 'your comparison is unfair.'*

Slide 21: Evaluation Pipeline Corrections

What is on the slide: Before vs final comparison: one-step approximation vs true autoregressive decoding, stochastic vs deterministic, BLEU bug risk fixed, logs added.

What to explain:

- Say the midterm version had evaluation weaknesses, and this final version fixes them.
- Before: baselines used one-step approximations and policy evaluation could be stochastic.
- Final: every system is decoded autoregressively, policy rollout is greedy/deterministic, and the SacreBLEU/chrF reference format is correct.
- Every prefix state is saved in JSONL/CSV logs.
- This slide is about credibility: the numbers are traceable and reproducible.

Speaker cue: *You can be direct here. It shows research maturity, not weakness.*

Slide 22: Logged Evidence per Source Prefix

What is on the slide: Table of logged fields: h(t), c(t), z(t), erasure, CNAL, g_t, action; bottom explanation about traceability.

What to explain:

- Explain that every aggregate metric comes from prefix-level rows.
- For each source prefix, we record the tentative hypothesis, committed prefix, visible output, action, erasure, and lag.
- This lets us inspect exactly where flicker occurred or where the policy delayed too much.
- For a paper, this is useful because it makes failure cases and ablations much more defensible.

Speaker cue: *This is an engineering/reproducibility slide. Keep it crisp.*

Slide 23: Main Result: WMT14 En-De Test Set

What is on the slide: Main result table comparing offline, immediate, wait-k, local agreement, confidence commit, and Predict-Then-Commit across BLEU, chrF++, COMET, DE, CNAL, and delay.

What to explain:

- Start with the offline full-sentence upper bound: BLEU 28.4.
- Immediate retranslation is strong in quality, BLEU 27.1, but has very high visible erasure: 31.8 characters per sentence.
- Wait-k reduces erasure as k increases but pays in latency and sometimes quality.
- Predict-Then-Commit reaches BLEU 26.9, chrF++ 55.8, COMET 0.751, with zero committed visible retraction.
- Key interpretation: we preserve about 94.7% of offline BLEU while eliminating committed visible retraction and keeping latency competitive.

Speaker cue: *This is the money slide. Speak slowly and compare rows, not every number.*

Slide 24: Quality-Latency-Stability Pareto Frontier

What is on the slide: Scatter plot: CNAL latency on x-axis, BLEU on y-axis; labels for Immediate, W1, W3, W5, LA-2, Predict-Then-Commit.

What to explain:

- Explain axes first: left is lower latency, up is better BLEU.
- Immediate is high quality and low CNAL, but it has high erasure, which is not shown directly by marker size here.
- Wait-k trades latency for stability as k increases.

- Predict-Then-Commit stays near the high-quality/low-latency region while also being the only zero-committed-erasure point.
- The takeaway is that we optimize the frontier, not one scalar metric.

Speaker cue: *Mention the slide note: marker labels omit erasure, but PTC has zero committed DE.*

Slide 25: Visible Stability Result

What is on the slide: Bar chart of average Detokenized Erasure per sentence: Immediate 31.8, W1 9.6, W3 5.7, W5 2.9, LocalAgree 1.6, ConfCommit 1.2, PTC 0.

What to explain:

- This slide isolates the stability objective.
- Immediate retranslation has the most flicker because it keeps changing the visible string.
- Wait-k improves stability gradually but still permits some visible revision.
- Local agreement and confidence commit reduce flicker further, but not fully.
- Predict-Then-Commit reaches zero committed visible erasure because it separates internal uncertainty from committed output.

Speaker cue: *This is the cleanest visual evidence for the core claim.*

Slide 26: IWSLT-Style TED En-De Generalization

What is on the slide: Generalization table on TED-style English-German: PTC with BLEU 25.5, chrF++ 53.8, COMET 0.724, DE 0.0, CNAL 8.6.

What to explain:

- Say this tests whether the learned policy only works on news-style WMT text or transfers to speech-like TED text.
- Immediate still has good quality but high flicker.
- Predict-Then-Commit keeps quality close to immediate while driving DE to zero.
- This supports generalization: the policy is not merely overfit to one dataset style.
- Keep the claim modest: it transfers beyond news-style text, not that it solves all domains.

Speaker cue: *This slide strengthens the paper angle.*

Slide 27: Ablation: Reward Components

What is on the slide: Table showing policy variants without erasure, latency, or commit-delay penalties, token-LCP erasure only, and balanced full reward.

What to explain:

- Explain why ablations matter: they show which part of the reward is doing what.
- Without erasure penalty, BLEU is high but DE is terrible: the model speaks too freely and flickers.
- Without latency or commit-delay penalties, DE can be zero, but the system waits too long.
- Token-LCP erasure is weaker because it does not perfectly match visible character-level changes.
- Balanced full reward gives the best practical tradeoff.

Speaker cue: *Key line: ‘Erasure penalty alone teaches silence; the full reward teaches stable speech.’*

Slide 28: Ablation: What Does the Policy Learn?

What is on the slide: Plot of P(COMMIT) versus source revealed percentage; behavioral interpretation bullets.

What to explain:

- Describe the curve: early in the sentence, commit probability is low, so the policy mostly reads.
- Around mid-sentence, the probability rises as more context resolves ambiguity.
- Near full context, the policy flushes remaining output quickly.
- This behavior matches intuition: high entropy means delay; low entropy plus mass balance means commit.
- This shows the policy is not just a fixed wait-k rule; it adapts across the sentence.

Speaker cue: *This is your interpretability slide.*

Slide 29: Metric Validation: DE Matches Visible Flicker Better Than Token LCP

What is on the slide: Table of correlations between metric predictors and user-visible flicker: token-LCP, word-level rewrite, DE, DE+CNAL.

What to explain:

- Explain the validation goal: prove DE is not just a pretty formula, but better aligned with visible flicker.
- Token-level LCP has weaker correlation because BPE boundaries distort what the user actually sees.
- Word-level rewrite count improves but still misses subword display effects.
- Detokenized Erasure correlates better because it measures surface character changes.
- DE plus CNAL performs best because it captures both instability and timing.

Speaker cue: *Use careful wording: say 'better proxy' rather than 'perfect human metric.'*

Slide 30: Case Study: Compound Noun Stress Test

What is on the slide: Qualitative example with source sentence about central bank interest rate decision; visible behavior and DE across systems; CNAL note.

What to explain:

- Use this as the concrete qualitative proof.
- The sentence stresses English-to-German compounding: 'central bank interest rate decision' can become a long German compound.
- Immediate retranslation visibly jumps through different interpretations, causing high DE.
- Wait-k and local agreement reduce but do not eliminate compound-boundary revisions.
- Predict-Then-Commit waits until the relevant ambiguity is resolved, then commits without retraction.
- CNAL helps because a long German compound carries high character mass even if it is one word.

Speaker cue: *This slide makes CNAL feel necessary, not decorative.*

Slide 31: Runtime and Deployment Feasibility

What is on the slide: Table with ms/update, GPU memory, prefix logs, real-time capable status; PTC 21.4 ms/update and 3.9GB.

What to explain:

- Explain that a streaming method must be deployable, not just accurate in a notebook.

- Immediate retranslation is expensive because it repeatedly regenerates full prefixes.
- Predict-Then-Commit is close to wait-k runtime while adding stability optimization.
- The character mass lookup is O(1) on token IDs, so CNAL accounting does not require expensive detokenization during policy rollout.
- This makes the method plausible for real-time use.

Speaker cue: *Do not oversell production readiness; say 'deployment-feasible prototype.'*

Slide 32: What Actually Worked

What is on the slide: Five reflection bullets: separation of h/c/z, DE, CNAL, MLE warm-start, frozen translator + lightweight policy.

What to explain:

- Use this as a synthesis slide.
- The biggest conceptual fix was separating tentative hypothesis, committed prefix, and visible output.
- The biggest metric fix was measuring display-stream erasure, not internal decoder changes.
- CNAL made latency more robust to tokenization and compounding artifacts.
- MLE warm-start and frozen translator made training stable by preventing all-READ collapse and catastrophic forgetting.
- This slide says what we learned technically.

Speaker cue: *Good transition into limitations.*

Slide 33: Failure Modes

What is on the slide: Table of failure modes, causes, and mitigations: German verb-final clauses, named entities, conservative policy, compounds, domain shift.

What to explain:

- Say that the method works well but not magically everywhere, because apparently syntax still exists to ruin our lives.
- Late verb-final German clauses remain hard because critical information may arrive near the end.
- Named entities can be ambiguous early without entity-type features.
- If λE is too high, the policy becomes over-conservative; commit-delay penalty helps.
- Domain shift can be handled with small-domain policy fine-tuning.
- This honesty makes the project more conference-ready.

Speaker cue: *Limitations should sound controlled, not fatal.*

Slide 34: Reviewer Questions We Are Ready For

What is on the slide: Three Q&A blocks: zero erasure trivial, fair baselines, why not local agreement.

What to explain:

- Answer Q1 first: zero erasure is trivial only if the system waits forever, so we report delay, CNAL, AL/DAL, and quality.
- Answer Q2: baselines are fair because all use true autoregressive decoding, identical tokenizer/test set, and deterministic evaluation.
- Answer Q3: local agreement is a strong heuristic, but it does not directly optimize the quality-latency-stability reward.

- This slide signals that the work has been stress-tested against obvious objections.

Speaker cue: *This is a confidence slide. No defensive tone.*

Slide 35: Positioning for A-Level Workshop / Conference Paper

What is on the slide: Paper angle, best-fit venues, and core sell: user-visible stability as missing objective, not 'RL for MT.'

What to explain:

- Say the project's strongest paper angle is not generic RL.
- The real contribution is formalizing user-visible instability and optimizing stable prefix commitment.
- The metric contributions are DE and CNAL; the policy contribution is learned stable-prefix commitment.
- Best near-term venues are WMT/IWSLT-style MT workshops, with ACL/EMNLP short paper potential after full validation.
- Core sell: this is user-visible stability as a missing objective in streaming translation.

Speaker cue: *This slide is for professor/advisor confidence.*

Slide 36: Contributions

What is on the slide: Five final contributions: stable-prefix formulation, DE, CNAL, policy-gradient commit model, corrected evaluation.

What to explain:

- Summarize the paper contributions clearly.
- First: a formal separation between h(t), c(t), and z(t).
- Second: DE measures visible text retraction at character level.
- Third: CNAL measures character-mass latency under subword tokenization.
- Fourth: a policy-gradient model learns READ/COMMIT behavior.
- Fifth: evaluation is corrected with proper metrics, deterministic rollout, true autoregressive baselines, and prefix logs.

Speaker cue: *This is the final technical summary before the closing message.*

Slide 37: Final Takeaway

What is on the slide: Closing statement: streaming translation should be stable enough to read; quality/latency/stability metric families shown.

What to explain:

- Deliver this slowly: 'Streaming translation should not merely be fast. It should be stable enough to read.'
- Restate that Predict-Then-Commit turns visible text retraction into something measurable and optimizable.
- Mention the three families one final time: quality, latency, and stability.
- End the main talk here unless the audience asks for appendix details.

Speaker cue: *This is your memorable ending. Pause after the first two sentences.*

Slide 38: Appendix A: Actor-Critic Extension

What is on the slide: Optional appendix defining value head, advantage, actor loss, critic loss, and warning about terminology.

What to explain:

- Use this only if someone asks why the title no longer says actor-critic or how actor-critic could be added.
- Explain that actor-critic requires a learned value function $V\psi(s_t)$.
- The current implementation is policy-gradient; adding the critic would replace the moving-average baseline with state-dependent value estimation.
- This is future extension material, not part of the current claim.

Speaker cue: *Appendix slide. Keep it technical and brief.*

Slide 39: Appendix B: Why SacreBLEU Reference Shape Matters

What is on the slide: Correct and incorrect corpus-level SacreBLEU formats and consequence of reference-format bug.

What to explain:

- Use this if asked about the evaluation correction.
- Explain that SacreBLEU expects predictions and a list of reference streams: predictions, [references].
- Per-sentence nesting changes the structure and can produce misleading scores.
- This bug is small in code but huge in credibility.

Speaker cue: *This slide helps justify why the final evaluation is stronger than the midterm version.*

Slide 40: Appendix C: Prefix Log Schema

What is on the slide: Schema table: sent_id, strategy, prefix_step, source_prefix, h_t, c_t, z_t, step_erasure, step_cnal, action.

What to explain:

- Use this if asked how the logs are structured.
- Each row corresponds to a streaming prefix event.
- The important fields are h_t, c_t, z_t, action, erasure, and CNAL.
- This schema makes aggregate metrics auditable back to individual visible output changes.

Speaker cue: *Good for technical audience questions.*

Slide 41: Appendix D: Hyperparameters

What is on the slide: Table of model, optimizer, learning rates, reward weights, and deterministic evaluation.

What to explain:

- Use this if someone asks implementation details.
- Mention the model is a Transformer/Marian-style BPE setup with 512 hidden size, 6 encoder/decoder layers, and 8 heads.
- Stage 1 uses AdamW at 1e-4; Stage 2 trains the policy head only at 1e-5.
- Reward weights balance erasure, latency, and delay.
- Evaluation is deterministic greedy decoding.

Speaker cue: *Do not spend time here during the main talk unless asked.*

Slide 42: Thank You

What is on the slide: Final thank-you slide with project title and quality-latency-stability tags.

What to explain:

- Thank the audience.
- If there is time, restate the one-line contribution: ‘We make visible text stability a measurable and optimizable objective for streaming MT.’
- Then invite questions verbally, not with a long rambling ending.
- Have slides 38-41 ready for technical questions.

Speaker cue: *End cleanly. Do not keep talking just because silence feels awkward. Humans do that, and it never helps.*

Recommended Main-Talk Flow

Section	Slides	Time	Purpose
Opening and motivation	Slides 1-6	2.5-3.5 min	Make the audience care about visible instability.
Formalization	Slides 7-14	4-5 min	Explain h(t), c(t), z(t), DE, CNAL, commit delay.
Method	Slides 15-19	3-4 min	Show the corrected system and training pipeline.
Evaluation protocol	Slides 20-22	2-3 min	Defend baseline fairness and logging.
Results and analysis	Slides 23-31	5-7 min	Focus on the table, Pareto frontier, stability, ablations, and case study.
Takeaways and positioning	Slides 32-37	2.5-3.5 min	End with limitations, reviewer answers, contributions, and final takeaway.
Backup	Slides 38-41	Only if asked	Use for actor-critic, SacreBLEU, log schema, hyperparameters.

Final one-line thesis to memorize: Streaming translation should not merely be fast; it should be stable enough to read.